

# Literature Status Note: Problem U in Totally Smooth Renormings

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## Status

**Status: literature\_already\_answered.** The Problem U and Problem U convex from Oja–Viil–Werner, *Totally smooth renormings*, arXiv:1807.07335, are answered positively in full generality by Cobollo–Guirao–Montesinos, *A remark on totally smooth renormings*, arXiv:2305.11611.

## Source Question

The source paper is Eve Oja, Tauri Viil, and Dirk Werner, *Totally smooth renormings*, arXiv:1807.07335, Archiv der Mathematik 112 (2019), 269–281.

In the introduction, after recalling that an  $M$ -ideal in its bidual has property  $U$ , the authors ask:

If  $X$  has property  $U$  in its bidual  $X^{**}$ , then does  $X$  admit an equivalent norm under which  $X$  is totally smooth in its bidual?

They label this as Problem U. They then use the Taylor–Foguel characterization to reformulate it as Problem U convex:

If  $X$  has property  $U$  in its bidual  $X^{**}$ , then does  $X$  admit an equivalent norm under which the dual space  $X^*$  is strictly convex and  $X$  still has property  $U$  in its bidual?

The same paper proves separable and WCG cases, with Theorem 3.3 giving the WCG partial answer.

## Supporting Answer

The supporting paper is Ch. Cobollo, A. J. Guirao, and V. Montesinos, *A remark on totally smooth renormings*, arXiv:2305.11611. Its abstract explicitly cites Oja–Viil–Werner and states that their WCG renorming result holds “in full generality” and in a stronger form, using a theorem of M. Raja on dual locally uniformly rotund norms.

The decisive result is Theorem 1 of arXiv:2305.11611. It proves that, for a Banach space  $X$ , the following conditions are equivalent:

1.  $X$  has an equivalent norm with Hahn–Banach smoothness (HBS), i.e. property  $U$  in the bidual.
2.  $X$  has an equivalent norm whose dual norm has the  $WW^*$ Kadets property.

3.  $X$  has an equivalent norm whose dual norm is LUR.
4.  $X$  has an equivalent norm with total smoothness.

The proof combines the Godefroy/Harmand–Werner–Werner equivalence between HBS and coincidence of the weak and weak-star topologies on the dual sphere, Raja’s equivalence between a dual LUR renorming and an equivalent  $WW^*$ Kadets norm, and the Taylor–Foguel theorem relating total smoothness to HBS plus rotundity of the dual norm.

## Why This Answers Problem U

Assume  $X$  has property  $U$  in  $X^{**}$  under its original norm. Then  $X$  already satisfies condition (i) of Theorem 1 in arXiv:2305.11611. Therefore it has an equivalent norm whose dual norm is LUR. In particular the dual norm is strictly convex. The same theorem also gives an equivalent norm with total smoothness. Thus Problem U has a positive answer in full generality.

Moreover, because the LUR dual renorming gives the strict convexity requested in Problem U convex while preserving the relevant Hahn–Banach smoothness/property  $U$ , the equivalent reformulation is also answered positively.

## Search And Classification Notes

The local cheap indexes had no prior packet for arXiv:1807.07335, Problem U, or the title *Totally smooth renormings*. Later local source arXiv:2506.08458 pointed to the same Raja dual-LUR mechanism, and an external arXiv search located arXiv:2305.11611, whose abstract and Theorem 1 explicitly identify the Oja–Viil–Werner problem. Since the supporting authors knew they were strengthening the Oja–Viil–Werner result, the correct classification is `literature_already_answered`.

## References

- [1] E. Oja, T. Viil, and D. Werner, *Totally smooth renormings*, Archiv der Mathematik 112 (2019), 269–281; arXiv:1807.07335.
- [2] Ch. Cobollo, A. J. Guirao, and V. Montesinos, *A remark on totally smooth renormings*, arXiv:2305.11611.
- [3] M. Raja, *On dual locally uniformly rotund norms*, Israel Journal of Mathematics 129 (2002), 77–91.